AI BASICS

Logistic Regression

Foundation of binary classification algorithms

Predicting Categories, Not Numbers

Estimating probability for categorical class membership.

Yes / No

True / False

Spam / Inbox

Why Not Linear Regression?

Linear Regression

- → Continuous value outputs.
- → Unbounded scale ranges.
- → Extreme outlier sensitivity.

Logistic Regression

- ✓ Categorical class prediction.
- ✓ Bounded probability boundaries.
- ✓ Mathematical output constraints.

STEP 1: THE FOUNDATION

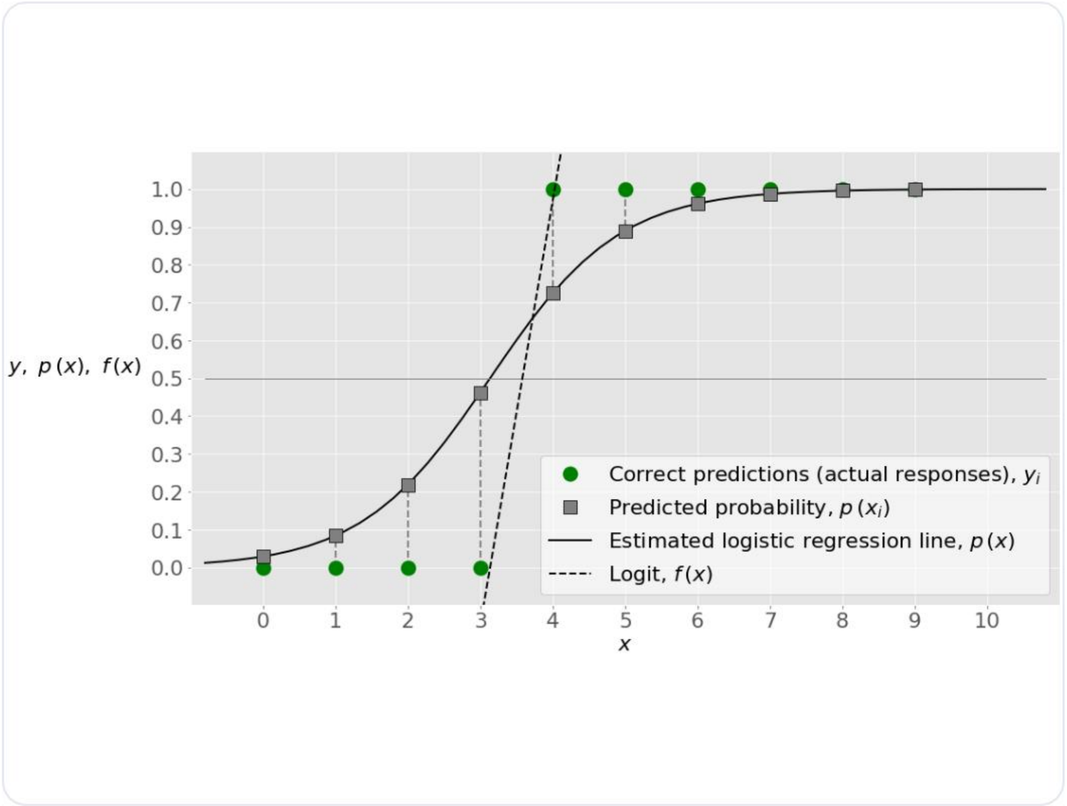
Linear Combination

$$z = w^T x + b$$

w = Feature weights **x** = Input features **b** = Bias term

STEP 2: THE TRANSFORMATION

The Sigmoid Function



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Maps real numbers to (0, 1).
- Characteristic S-shaped gradient.
- Outputs probability estimations.

STEP 3: THE OUTPUT

Decision Boundaries

$$P \geq 0.5$$

Positive class prediction.

$$P < 0.5$$

Negative class prediction.

MODEL EVALUATION

Binary Cross-Entropy

$$L = -\left[y \log\left(\hat{y}\right) + \left(1 - y\right) \log\left(1 - \hat{y}\right) \right]$$

Heavily penalizes highly confident incorrect predictions.

APPLICATION EXAMPLE

Email Spam Classification

Sender Domain

Trust metrics evaluation.

Keyword Frequencies

Suspicious terminology counts.

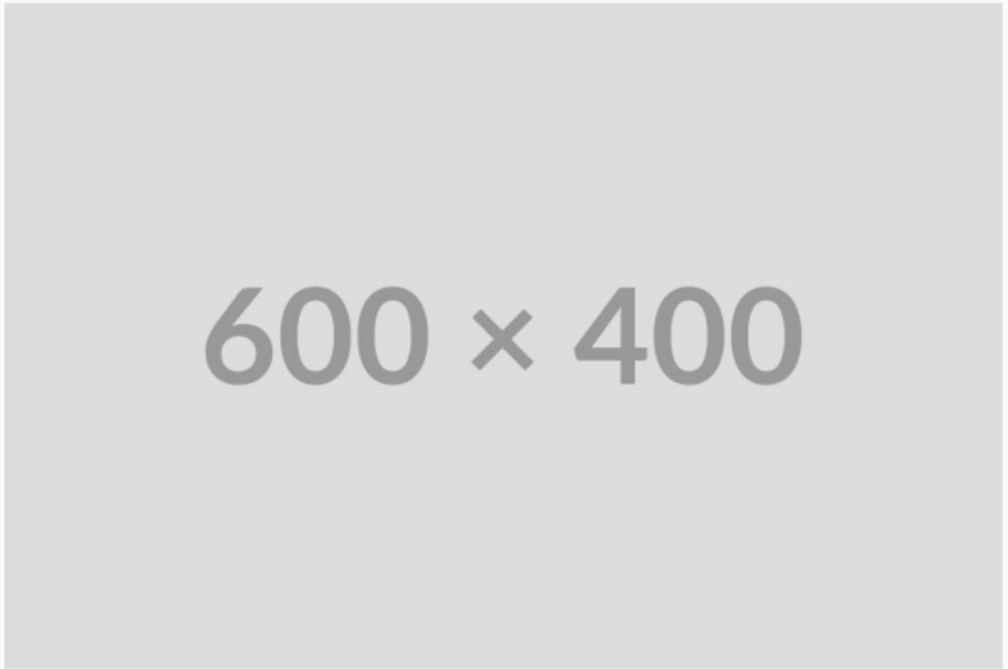
Output Execution

Automatic inbox routing.



APPLICATION EXAMPLE

Disease Risk Diagnosis



Biomarker Levels

Blood pressure metrics.

Patient Demographics

Age and BMI statistics.

Clinical Support

Statistical risk probability.

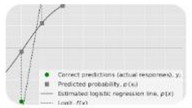
Summary & Questions

Interpretable, efficient probabilistic classification.

KEY TAKEAWAYS

- Probability based outputs.
- Sigmoid curve transformation.
- Binary decision thresholds.

Image Sources



https://files.realpython.com/media/log-reg-7.9141027bd736.png

Source: realpython.com



https://www.mailpro.com/blog/img/aiimagegenerator.jpg

Source: www.mailpro.com